

Algorithm for file updates in Python

Project description

This project is showcasing how to define an algorithm to update a file in Python. We'll be using the Jupyter Notebook as an IDE.

Open the file that contains the allow list

The file we want to open is called `"allow_list.txt"`. We'll assign a string containing this file name to the `import_file` variable

We'll create a `list` to include the IP addresses that we want to remove from the file. We'll assign the IP addresses into the `remove_list` variable.

Now, we'll use a `with` statement to open it. Use the variable `file` to store the file while you work with it inside the `with` statement.

```
# Assign `import_file` to the name of the file
import_file = "allow_list.txt"

# Assign `remove_list` to a List of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]
|
# First line of `with` statement
with open(import_file, "r") as file:
```

Read the file contents

Next, we'll use the read method to convert the contents of the allow list file into a string. We'll store this string into a variable called `ip_addresses`.

```
In [ ]: # Assign `import_file` to the name of the file
import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file
with open(import_file, "r") as file:

    # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`
    ip_addresses = file.read()

# Display `ip_addresses`
print(ip_addresses)
```

Convert the string into a list

To remove the IP addresses from the allow list, they need to be in a list format. For this purpose we'll use the `.split()` method to convert the `ip_addresses` into a list.

```
In [ ]: # Assign `import_file` to the name of the file
import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file
with open(import_file, "r") as file:

    # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`
    ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list
ip_addresses = ip_addresses.split()

# Display `ip_addresses`
print(ip_addresses)
```

Iterate through the remove list

A second list called `remove_list` contains all of the IP addresses that should be removed from the `ip_addresses` list. We'll set up the header of a `for` loop that will iterate through the `remove_list` using `element` as the loop variable.

```

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file
with open(import_file, "r") as file:
    # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`
    ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list
ip_addresses = ip_addresses.split()

# Build iterative statement
# Name loop variable `element`
# Loop through `ip_addresses`
for element in ip_addresses:
    # Display `element` in every iteration
    print(element)

```

Remove IP addresses that are on the remove list

In the body of the iterative statement we'll add code that will remove all the IP addresses from the allow list that are also on the remove list. In order to do that we'll need to create a conditional statement. First, we'll create a conditional statement that evaluates if the loop variable **element** is part of the **ip_addresses** list. Then, within that conditional, we'll apply the `.remove()` method to the `ip_addresses` list and remove the IP addresses identified in the loop variable `element`. Using the `.remove()` method is possible in this case because there are no duplicates in the **ip_addresses** list.

```

: # Assign `import_file` to the name of the file
import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file
with open(import_file, "r") as file:

    # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`
    ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list
ip_addresses = ip_addresses.split()

# Build iterative statement
# Name loop variable `element`
# Loop through `ip_addresses`
for element in ip_addresses:

    # Build conditional statement
    # If current element is in `remove_list`,
    if element in remove_list:

        # then current element should be removed from `ip_addresses`
        ip_addresses.remove(element)

# Display `ip_addresses`
print(ip_addresses)

```

Update the file with the revised list of IP addresses

We can complete the algorithm by updating the file with the revised list of IP addresses. In order to do this we must convert the `ip_addresses` list back into a string using the `.join()` method, and finally we'll rewrite the file using the `write()` method.

```

# Assign `import_file` to the name of the file
import_file = "allow_list.txt"

# Assign `remove_list` to a List of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file
with open(import_file, "r") as file:

    # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`
    ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list
ip_addresses = ip_addresses.split()

# Build iterative statement
# Name loop variable `element`
# Loop through `ip_addresses`
for element in ip_addresses:

    # Build conditional statement
    # If current element is in `remove_list`,
    if element in remove_list:

        # then current element should be removed from `ip_addresses`
        ip_addresses.remove(element)

# Convert `ip_addresses` back to a string so that it can be written into the text file
ip_addresses = " ".join(ip_addresses)

# Build `with` statement to rewrite the original file
with open(import_file, "w") as file:

    # Rewrite the file, replacing its contents with `ip_addresses`
    file.write(ip_addresses)

```

Summary

The project displayed creating an algorithm that removes IP addresses in a remove list variable from the allow_list.txt.